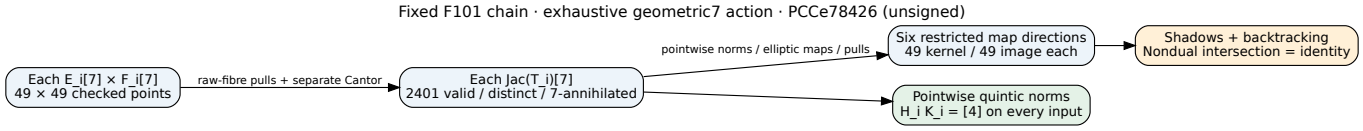
Complete geometric 7-torsion action

2026-11-12 · fixed F101 Jacobian chain · scientific standing unsigned

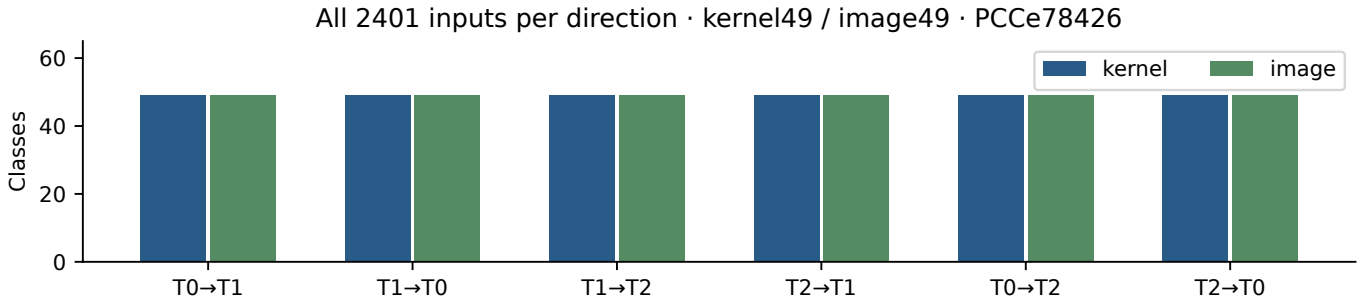
PCCe78426 records30 passing standalone tests,with zero failures,ignores or filters. Formatting,strict Clippy and the old reference/legacy guards pass. All four Rust sources and nine fixture hashes remain unchanged. The whole 22,566-byte raw stream is retained as standard XZ5,008 bytes,with raw and archive digests and exact roundtrip. These are correctness checks. [pcc:unverified]



Exhaustive source populations

For each of T0,T1,T2,the existing stable7 edges generate49 checked points in each elliptic factor. Their2,401 product pullbacks are valid,distinct, killed by7 and fixed by Frobenius^6. The independent pointwise norms in the reduced quintic chart equal four times every elliptic-product input. The branch-point base divisor,finite poles,multiplicities,ramification and signed original infinities are retained. [pcc:unverified]

The admitted identity $H_iK_i=[4]$ makes K_i injective on7-torsion. Milne’s prime-to-characteristic torsion theorem gives exactly$7^4=2{,}401$ geometric points on a genus2 Jacobian. Under the written Picard and group-law dependencies,the checked distinct population therefore exhausts $A_i[7]$. This is stronger than testing the previously selected49-class kernels. [pcc:unverified]



All six first,second and compressed forward/adjoint directions agree on every2,401 inputs with the direct elliptic-product route. Each restricted map has kernel49,image49,and every image is killed by7. The evaluator uses the existing field,polynomial,elliptic,Kohel and separate legacy Cantor engines;it does not import the producer Balanced,trace,Richelot or pipeline implementation. Shared arithmetic dependencies remain explicit. [pcc:unverified]

Chain orientation and degree controls

On the shared T1 source,the first adjoint kernel and second forward kernel intersect only in the identity. The compressed forward7-kernel equals the first forward kernel;the compressed reverse7-kernel equals the second adjoint kernel. Backtracking through either single hop and its actual adjoint kills all2,401 source classes. [pcc:unverified]

Checked control	Recorded outcome
All six directions:kernel / image	49 /49 on each2,401 source
Shared T1 kernel intersection	1 identity
Single-hop forward then adjoint	2,401 zeros for each hop
CT38 scalar4,976,069	All2,401 images become zero
CT38 scalar1,244,018	2,352 differences from the map per direction
Deeper scalar8,708,120	Identity on all7-torsion controls

Conditional full compressed kernel type

Admit the old global separable degree7^4 and norm49 identities. The full geometric compressed kernel L then has order7^4 and is killed by49. This census gives |L[7]|=49. Finite abelian classification writes $L=(Z/49)^r\times(Z/7)^s$,so2r+s=4 and r+s=2 give r=2,s=0. Thus the abstract kernel is $(Z/49)^2$ under those dependencies. The census does not enumerate the full49-torsion or the full kernel over $F101^{42}$. The earlier cyclic49 transport proof remains prior work. [pcc:unverified]

Why the shadow controls remain necessary

CT38′s first scalar is0 modulo7;the second is−1. Their restricted actions are rejected on every nonzero image. However8,708,120=1+7·101·12,317 acts as identity on the rational group,tangent module and7-torsion. Even this exhaustive finite census cannot replace a global degree proof. [pcc:unverified]

SevenCensusControls.nonzero_seven_shadow_rejected/a6ff17 and deeper_shadow_retains_finite_controls/90cebe faithfully formalize these module consequences with explicit annihilation hypotheses. Both are individually bound and locally pass;gate03 checks two statements with zero trusted native proofs. The written census is149 declarations across30 modules. Actual varieties,torsion realization,Picard maps,Cantor correctness and finite-group classification remain partial or written dependencies. Failed proof attempts01/1a2d21 and02/a77458 retain their complete sources, builds and gates;failed sorryAx output is never counted. [pcc:unverified]

Sources and current checkpoint

Written proof d89ae5 maps every conclusion to its dependencies. Milne, Abelian Varieties,Theorem7.2/ Remark7.3 supplies geometric torsion. External CT41 atca083268 and primary Howe–Lauter §2 are pinned by f07968;they motivate the next actual4-gluing test. KB tools/wrapper and Conductor are unavailable; local accepted proofs and primary papers were used. No novelty is claimed. [pcc:unverified]

Frozen root67b exactCI43205484930 succeeds25native/29standalone/147written 29modules/full57d865. Priorb351 exact43196578049 succeeds21/14428/full3b94c3. Consumer1508/c712 exact43202427032 is still running at the scheduled observation;its accepted local scope is25native/98reference plus five integrations. The new30/14930 root needs its own checkpoint and exact CI. Existing frozen figures keep their historical scopes;these figures show the added geometric census. [pcc:unverified]

Next:full elliptic4-torsion,ramified pullbacks and actual quartic gluing anti-isometry on the three T_i. Benchmark admission still lacks a live shared balance;the atlas vault is locked. Native correctness and proof work continues. [pcc:unverified]